

EDA

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2026-03-30

EDA of birth and fertility in Singapore from 1960-2024

Introduction & Research Questions

This is the EDA conducted on birth and fertility data from Singapore from 1960-2024. We will be looking at 2 data series' which were recorded annually: Total Live Births (TLB) & Total Fertility Rate (TFR).

1. How has the total fertility rate changed in Singapore from 1960 to 2024?
2. What is the relationship between Total Live Births and Total Fertility Rate and do they show the same temporal behavior?
3. Are these series stationary across the time period?

We can answer each of these research questions through a basic plot of the data over time and a comparison and plotting of each variable against each other. We will use a simple linear regression to achieve this.

Data Description & Cleaning

Load packages and data

```
pacman::p_load(fpp3, skimr, tsibble, dplyr, readr)
BFR <- read_csv("BirthsAndFertilityRatesAnnual.csv")
```

```
## Rows: 17 Columns: 67
## -- Column specification -----
## Delimiter: ","
## chr (22): DataSeries, 2025, 1979, 1978, 1977, 1976, 1975, 1974, 1973, 1972, ...
## dbl (45): 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017, 2016, 2015, 2014, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

The following code transforms the ingested data into a more usable format of just the rows and columns we need and changes the names to be easier to use.

```
BFR <- BFR |>
  filter(DataSeries %in% c("Total Fertility Rate (TFR)", "Total Live-Births")) |>
  mutate(DataSeries =
    recode(DataSeries,
      "Total Fertility Rate (TFR)" = "TFR",
      "Total Live-Births" = "TLB" )) |>
  select(DataSeries, `1960`:`2024`) |>
  mutate(across(`1960`:`2024`, as.numeric)) |>
  pivot_longer(
    cols = `1960`:`2024`,
    names_to = "Year",
```

```

    values_to = "Value") |>
  mutate(Year = as.integer(Year)) |>
  pivot_wider(
    names_from = DataSeries,
    values_from = Value ) |>
  as_tsibble(index = Year)

```

BFR

```

## # A tsibble: 65 x 3 [1Y]
##   Year    TFR    TLB
##   <int> <dbl> <dbl>
## 1  1960  5.76 61775
## 2  1961  5.41 59930
## 3  1962  5.21 58977
## 4  1963  5.16 59530
## 5  1964  4.97 58217
## 6  1965  4.66 55725
## 7  1966  4.46 54680
## 8  1967  3.91 50560
## 9  1968  3.53 47241
## 10 1969  3.22 44562
## # i 55 more rows

```

Now to check if there are any missing values or anything strange happening with the data.

```
glimpse(BFR)
```

```

## Rows: 65
## Columns: 3
## $ Year <int> 1960, 1961, 1962, 1963, 1964, 1965, 1966, 1967, 1968, 1969, 1970, ~
## $ TFR <dbl> 5.76, 5.41, 5.21, 5.16, 4.97, 4.66, 4.46, 3.91, 3.53, 3.22, 3.07, ~
## $ TLB <dbl> 61775, 59930, 58977, 59530, 58217, 55725, 54680, 50560, 47241, 44~

```

```
summary(BFR)
```

```

##      Year      TFR      TLB
## Min.   :1960   Min.   :0.970   Min.   :33541
## 1st Qu.:1976   1st Qu.:1.260   1st Qu.:39570
## Median :1992   Median :1.620   Median :42484
## Mean   :1992   Mean   :2.063   Mean   :44360
## 3rd Qu.:2008   3rd Qu.:2.070   3rd Qu.:48577
## Max.   :2024   Max.   :5.760   Max.   :61775

```

```
colSums(is.na(BFR))
```

```

## Year  TFR  TLB
##    0    0    0

```

The outputs show that there are 65 distinct observations for each year with no missing or out of the ordinary values in the data set.

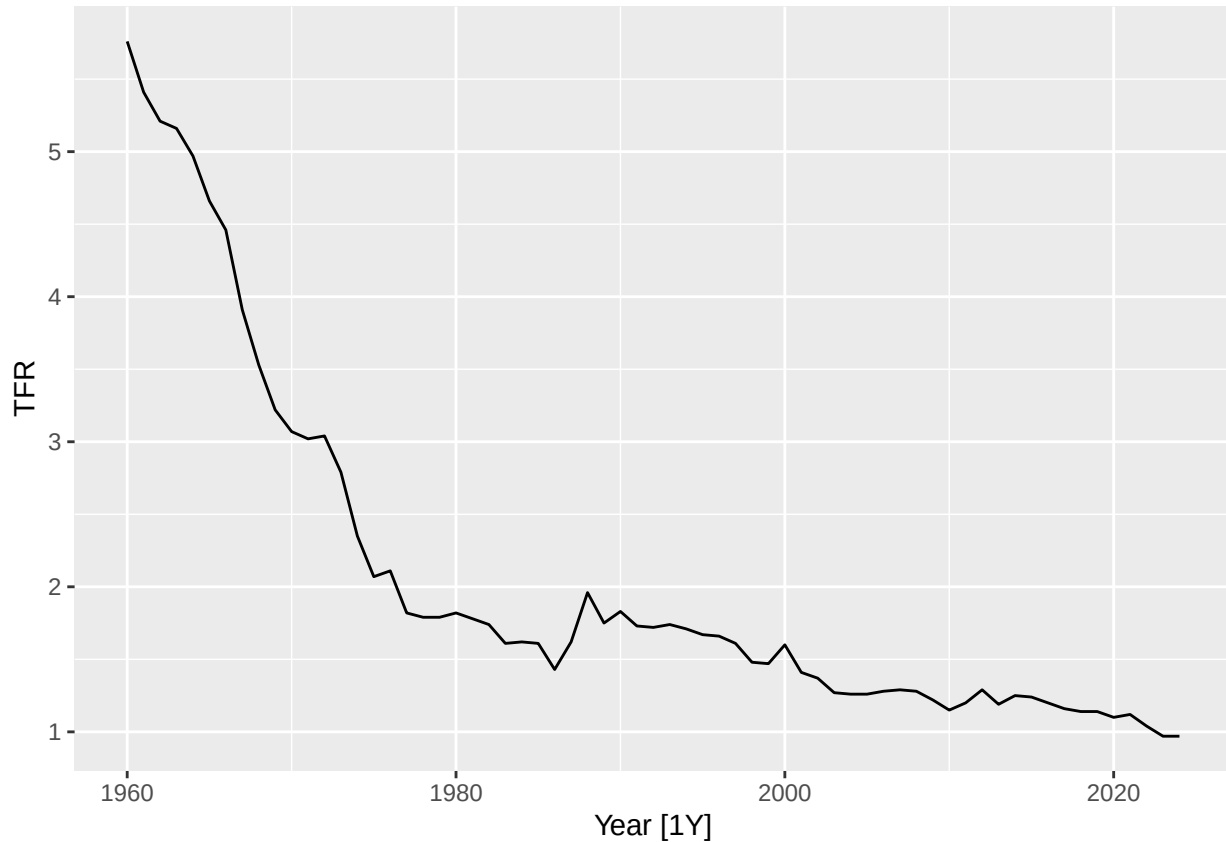
Since the data is annual we will not be able to see specific seasonality within each year.

Time Series Feature Analysis

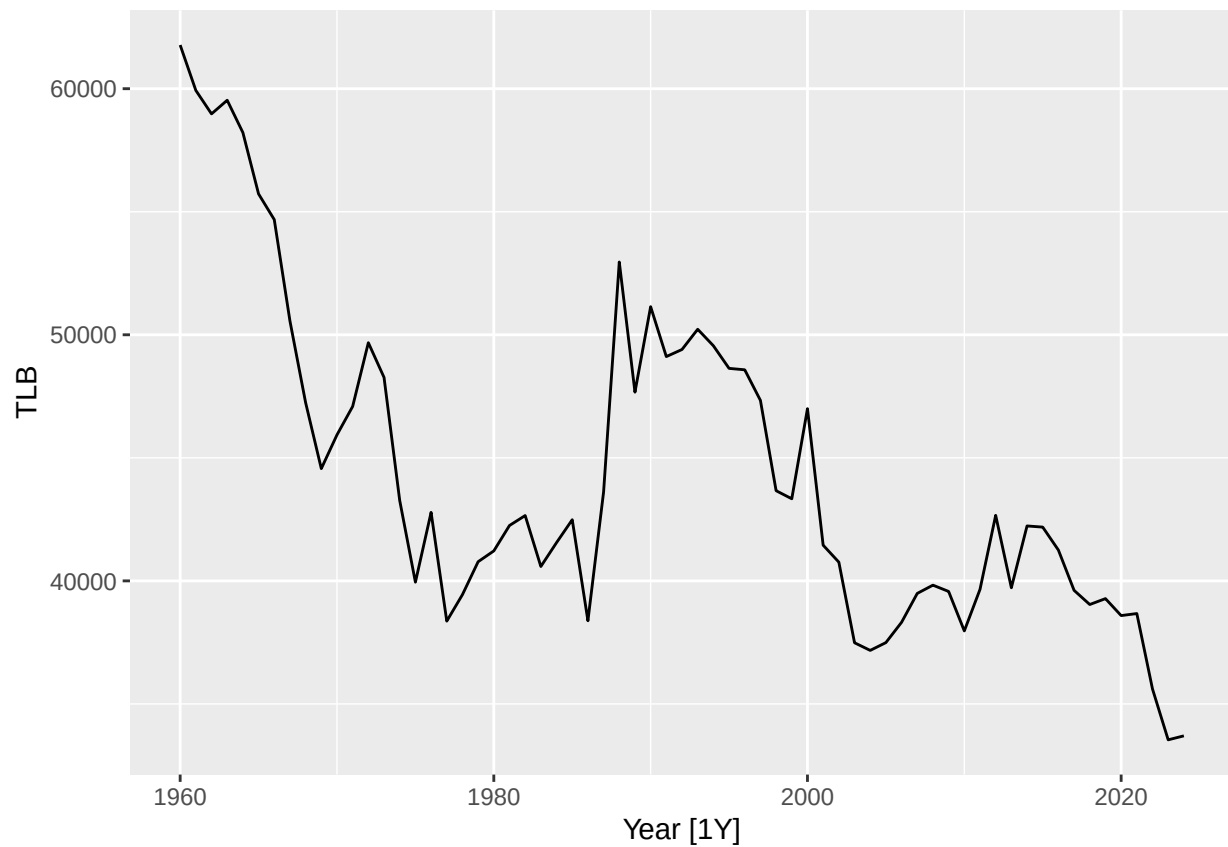
Here are some graphs of the TFR and TLB across the time period. Both TFR and TLB show decreasing trends since 1960. TFR's decrease is a very steep from its peak in 1960 to 1975 then it seems to level off at

a lower fertility rate. This will suggest non-stationarity in the mean. TLB also declines overall with more fluctuation than TFR suggesting an increase in overall population level at points throughout the series. The rate of decline across both TFR and TLB is similar at the start however the increase bump around 1985 is much stronger in TLB rather than the light bump we see in the fertility rate. They also have a different structure with TLB being more variable and TFR having a more consistent drop off after 1985.

```
BFR |>  
  autoplot(TFR)
```

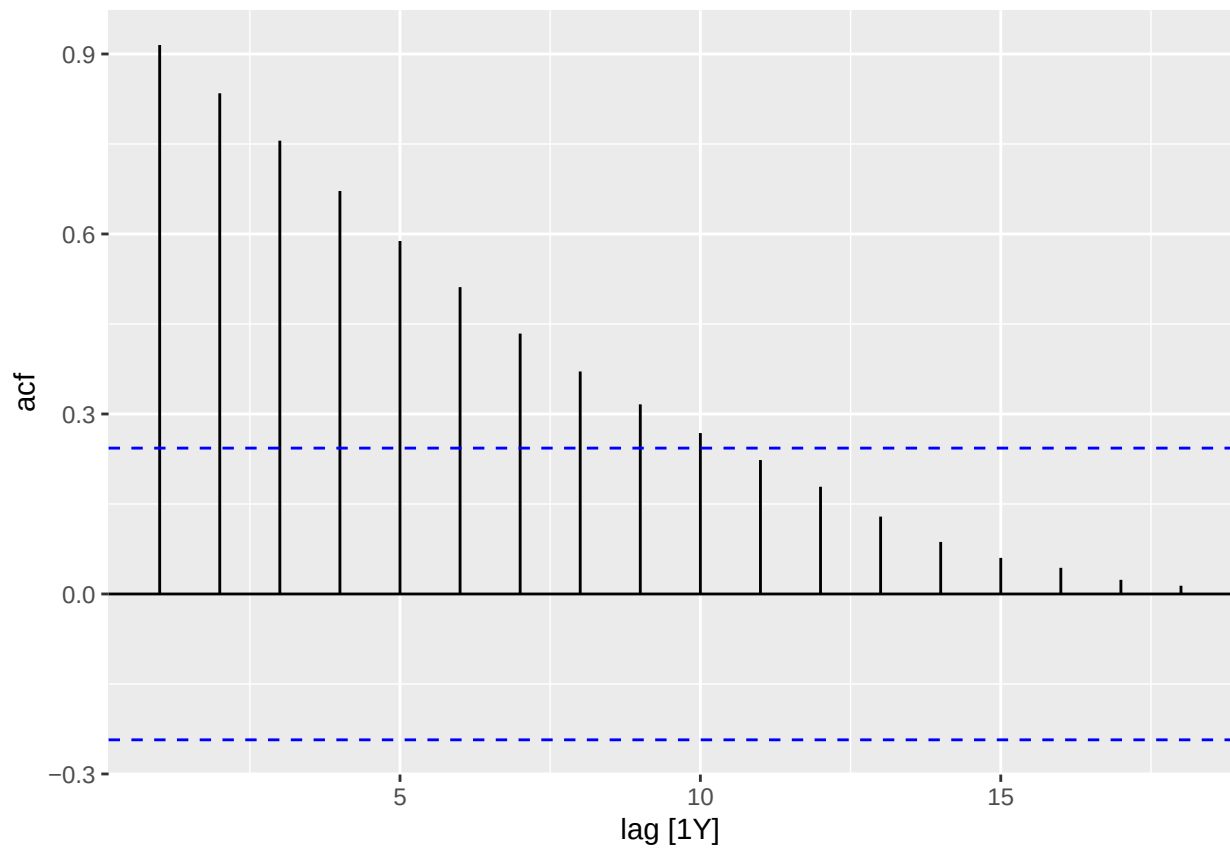


```
BFR |>  
  autoplot(TLB)
```

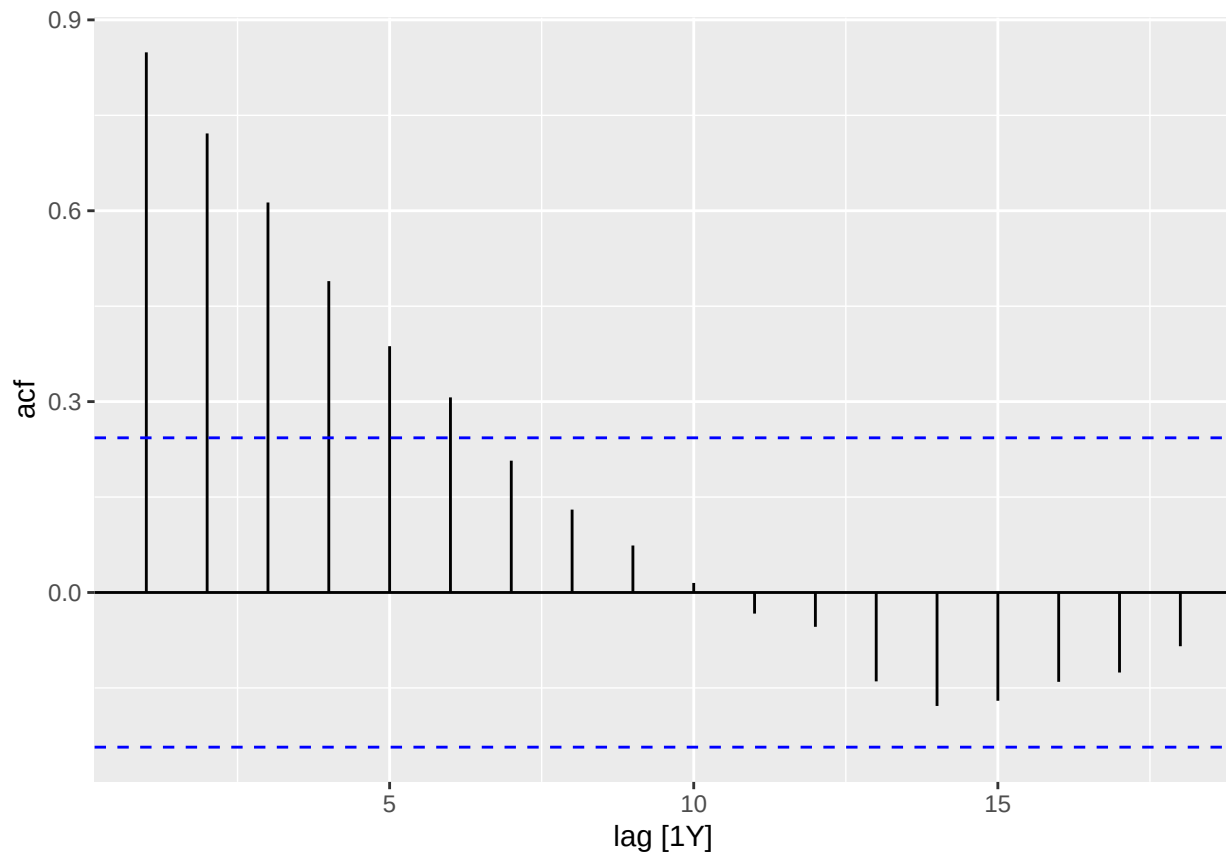


Both of the following ACF plots show a slow decay across lags. This supports the earlier theory of non-stationarity due to a changing mean.

```
BFR |>  
  ACF(TFR) |>  
  autoplot()
```



```
BFR |>  
  ACF(TLB) |>  
  autoplot()
```

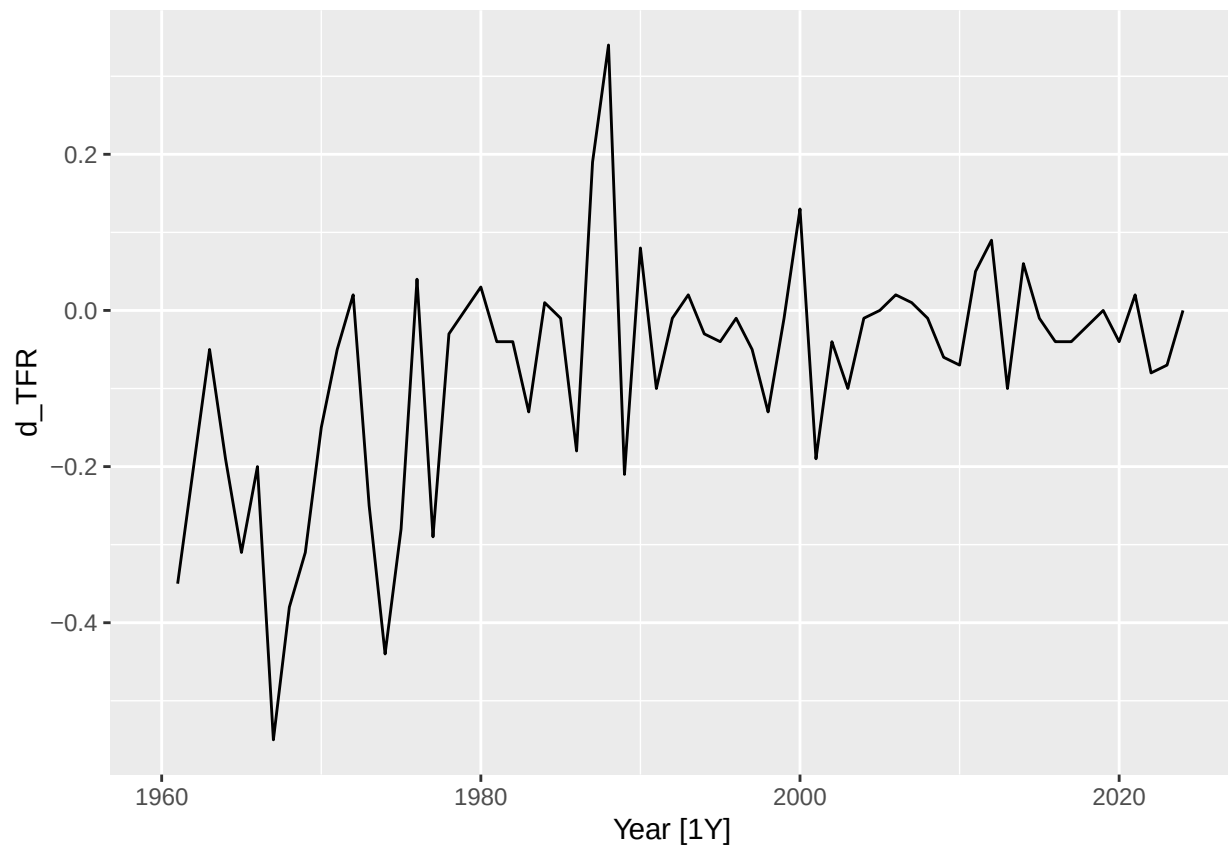


As discussed in section 1.5.1 of Introduction to Time Series Analysis, with an absence of seasonality to eliminate a trend and make the series stationary we have two options; to fit a polynomial trend and then subtract from the data or to apply differencing. Differencing is preferred since it does not assume a consistent form of the trend in the data. So to transform the data into a stationary series we will apply differencing.

```
BFR_diff <- BFR |>
  mutate(
    d_TFR = difference(TFR),
    d_TLB = difference(TLB)
  )
```

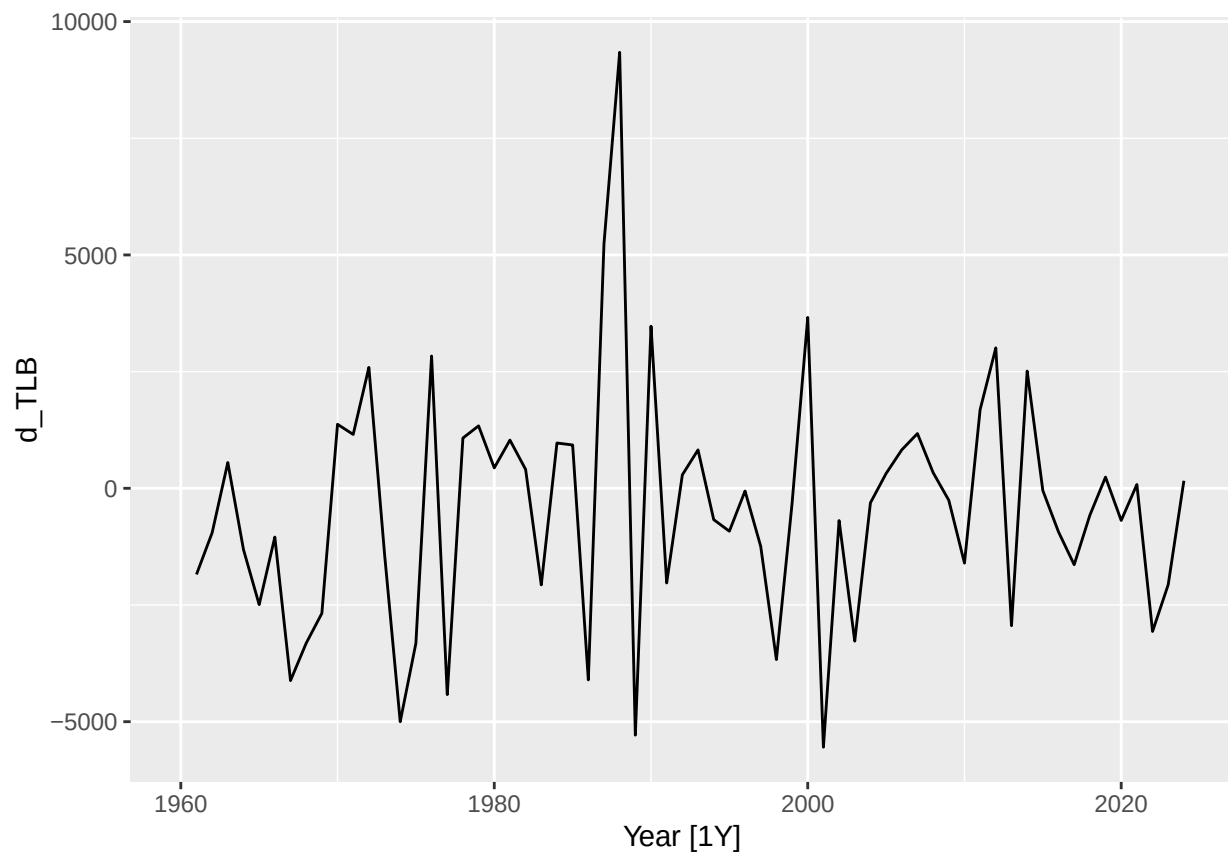
```
BFR_diff |>
  autoplot(d_TFR)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```



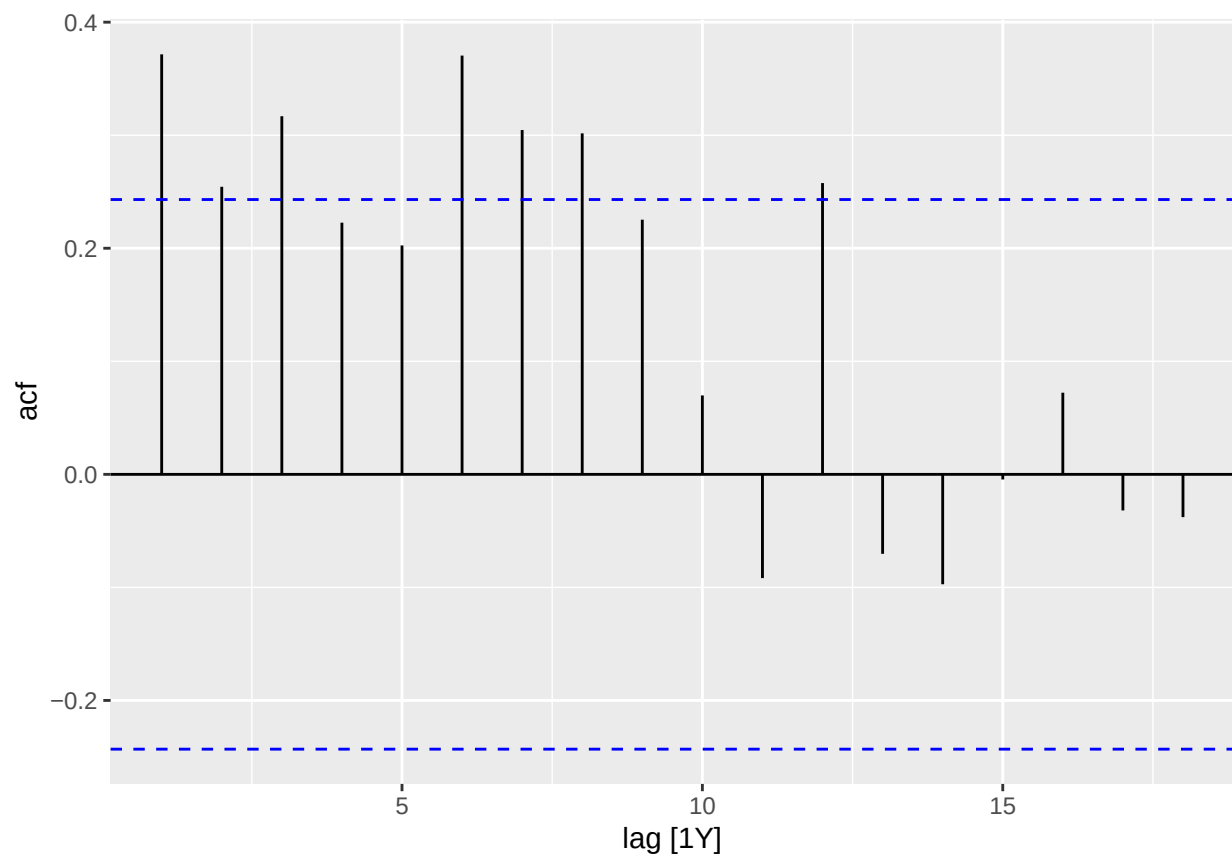
```
BFR_diff |>
  autoplot(d_TLB)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_line()`).
```

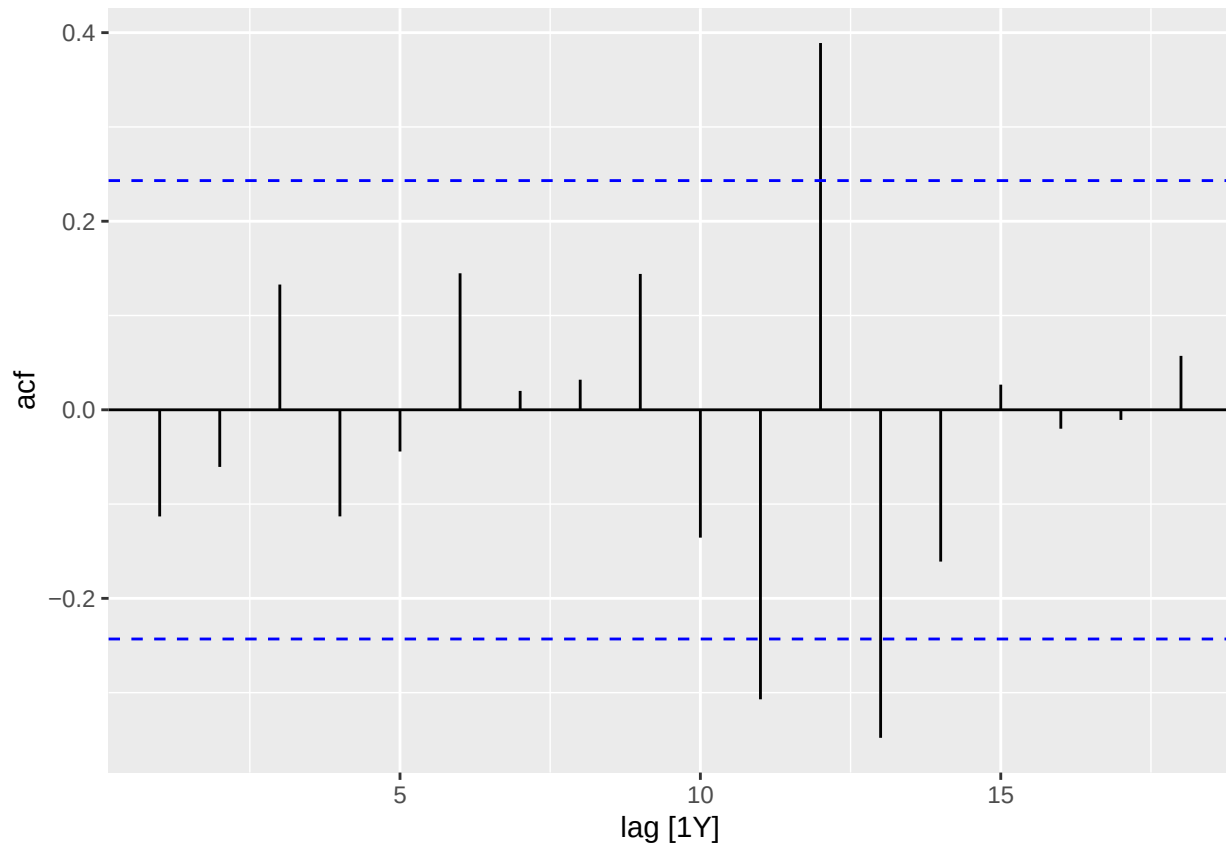


After applying first differencing we see that both TLB and TFR seem to fluctuate around a constant mean, this shows that the decreasing trend we saw earlier has been removed.

```
BFR_diff |>  
  ACF(d_TFR) |>  
  autoplot()
```

```
BFR_diff |>  
  ACF(d_TLB) |>  
  autoplot()
```



The auto correlation functions of the differenced series are much smaller and decrease much more rapidly than the previous iteration and so this also supports the stationarity of the data.

Time Series Features

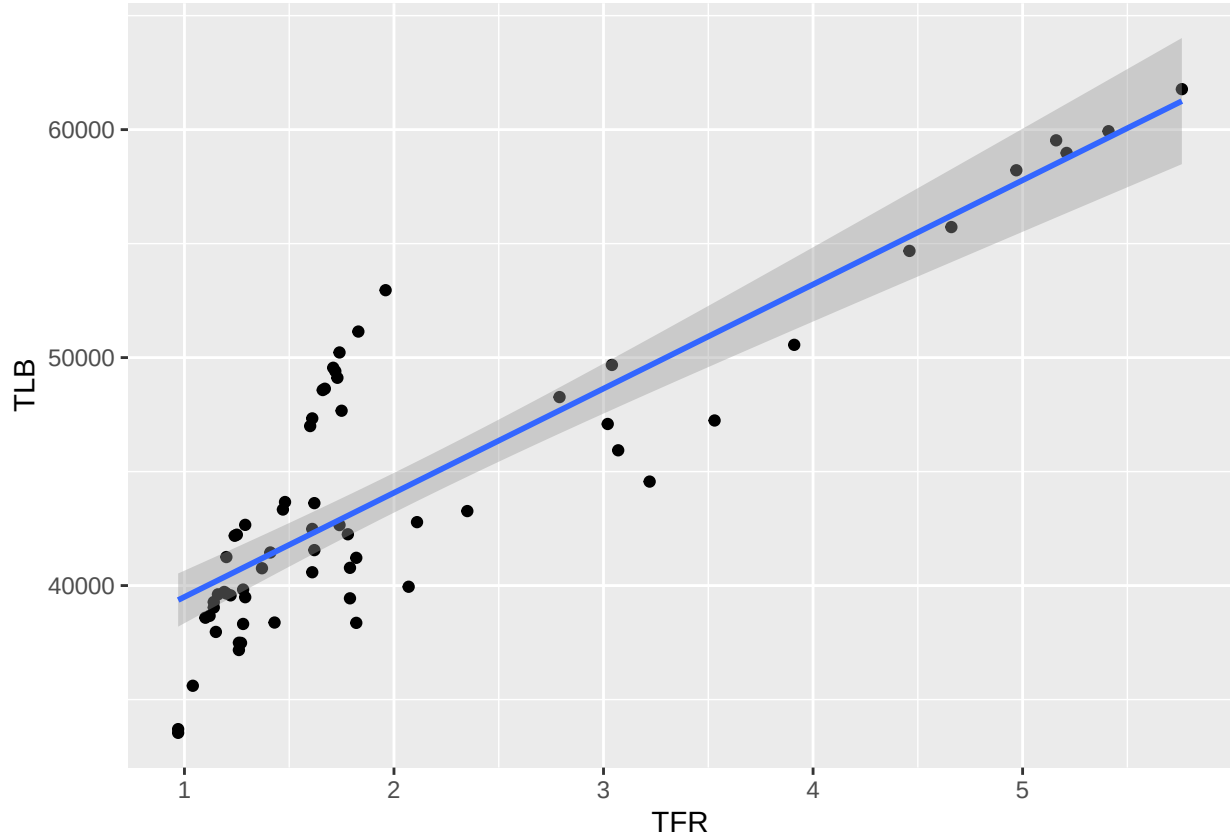
This series shows a strong downward trends, indicating a changing mean over time and as such non-stationarity. No seasonal patterns are seen due to the frequency of the data. The autocorrelation shows a strong persistence prior to differencing which removes much of the dependence.

Relationship of TFR and TLB

Below you will see a scatter plot of the relationship between TFR and TLB this is to answer the second research question about the relationship between our series. We will use a simple linear regression to fit a line to predict the relationship between TFR and TLB.

```
ggplot(BFR, aes(x = TFR, y = TLB)) +  
  geom_point() +  
  geom_smooth(method = "lm")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The trend in the scatterplot is positive. This says that higher fertility rates correlate to higher total live births. However, the spread of the data suggests that there are more factors to TLB than just fertility rate.

Limitations

This analysis has a couple of limitations due to the data source and its nature. The data comes in annually and so we cannot see within year trends or seasonality. The total live births are also influenced by other factors than TFR, such as population size and the age demographic of the population. As we are not looking at all the factors we cannot definitively interpret the complete nature of the relationship.

Summary of Findings

In summary both TFR and TLB showed signs of non-stationarity and downward trends from 1960-2024. The autocorrelation functions of both series shows temporal dependence. The first differencing removes the trend component of the data and therefore makes the data more stationary. There is a positive trend between TLB and TFR, however, there are more factors influencing TLB and they need to be considered as we build a prediction model for these time series.

Statistical Appendix

Stationarity: A time series is stationary if it has constance mean and variance across time.

ACF (Autocorrelation Function): Measures the correlation between each observation at different time lags

Differencing: A transformation of data defined as $\nabla X_t = X_t - X_{t-1} = (1 - B)X_t$ where B is the backward shift operator $BX_t = X_{t-1}$. This transformation aims to remove trends in non seasonal data and stabalise the mean across the series.

Lag: The number of fixed time displacement between an observation and a previous value in the dataset.

Simple Linear Regression: Is a model which is used to model the relationship between two variables. The model is defined as $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$; where Y_i is the response variable (TLB), X_i is the predictor variable (TFR), β_0 is the intercept value, β_1 is the slope coefficient representing the change in Y with a 1 unit increase in X , and ϵ is the error term assumed to have a mean of 0 and a constant variance across all observations.